Name	HARNESS,WIRING	Engineering Standard Number
Identifier	CORPORATE MANUFACTURING SPECIFICATION (ITEM)	18227

Abstract

This document specifies the required manufacturing techniques, some materials, assembly, test methods, and specification procedure and requirements required for wiring harness assemblies utilized in electrical/electronic systems used in Cummins Inc. products across all Business Units.

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1. Scope

This document specifies the required manufacturing techniques, some materials, assembly, test methods, and specification procedure and requirements required for wiring harness assemblies utilized in electrical/electronic systems.

2. Applicable Documents

Applicable documents listed below may be obtained from the respective organizations listed in CES 10054, Standards Organizations Addresses.

a. ANSI- J-Std-004, Requirements for Soldering Fluxes
b. CES 10054, Standards Organizations Addresses
c. CES 10056, Glossary
d. CES 10903, Substances, Prohibited and Restricted
e. CES 14311, Electronics Mechanical Requirements
f. CES 14232, Standard, Electronic Environmental
g. CES 18287, Barcode Labeling & Direct Part Marking
h. CES 19041, Global Packaging Std Production Parts
i. CES 20061, Grease, Mineral Oil
j. DOD-STD-1686, Electrostatic Discharge Control Program for Protection of Electrical and Electronic Parts, Assemblies and Equipment (Excluding Electrically Initiated Explosive Devices)
k. MIL-W-16878, Wire, Electrical, Insulated, General Specification For
l. O-E-760, Ethyl Alcohol (Ethanol); Denatured Alcohol; Proprietary Solvents and Special Industrial Solvents
m. SAE J1127, Low Voltage Battery Cable
n. SAE J1128, Low Voltage Primary Cable
o. SAE J1939, Recommended Practice for a Serial Control and Communications Vehicle Network
p. TT-I-735, Isopropyl Alcohol
q. ISO 6722-1, Road Vehicles - 60 V and 600 V Single-Core Cables-Part 1: Dimensions, Test Methods and Requirements for Copper Conductor Cables

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3. Definitions

Terms used in this standard that have a general definition for usage in Cummins Engineering Standards are defined in CES 10056, Glossary.

3.1. Cable Scanner

A programmable device that automatically checks electrical and electronic cables for open circuits, short circuits and point-to-point circuits.

3.2. Connector Backshell

A connector backshell is a rigid accessory that attaches to the rear of a connector housing. The primary purpose of the backshell may be either for cosmetic dressing, wire routing, wire support, shield from high pressure spray, or stress relief of the wires. Cosmetic dressing is secondary.

3.3. Dual Wall Heat Shrink Tube

Dual wall Heat Shrink Tube consist of a crosslinked outer jacket and an inner layer of adhesive or encapsulant.

3.4. Manifold

A rigid hinged fitting used to aid in harness routing. Available in various outlet configurations and conduit sizes to accommodate various routing options and configurations.

3.5. 90° Elbow (Manifold)

Hinged manifold with two openings offset by 90° which can be used to drive more rigid and controlled angles when routing bundles.

3.6. T-Joint (Manifold)

Hinged manifold with three openings offset by 90° which can be used to drive more rigid and controlled angles at junctions.

3.7. Y-Joint (Manifold)

A fitting with three openings, a wye joins (or creates) branch lines.

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4. Tools and Equipment

4.1. Tool Selection

Tools shall be suitable for the purpose intended and shall not cause damage to the parts. The following tools are considered suitable for their application. Tools and equipment shall be maintained and calibrated as specified by their manufacturer according to the harness supplier's Manufacturing Process Control Plan and, as needed, agreement with Cummins.

4.2. Mechanical Strippers

Mechanical strippers utilized to remove insulation from stranded or solid conductor wire may be of the hand operated or automatic high volume machine type. Hand operated strippers shall be of a fixed die configuration. Automatic high volume machine strippers shall be of a type utilizing either fixed dies, dies adjustable to calibrated stops, or roller cutters adjustable to calibrated stops. Dies, whether adjustable or fixed, shall be properly maintained to assure consistently sharp and even cuts without damage to the wires or unstripped insulation.

4.3. Thermal Stripper

Thermal strippers utilized to remove insulation from stranded and solid conductor wires shall be of a type that can be regulated to provide the required temperature. Temperature controls shall be sufficient to prevent damage to the wire or unstripped insulation. (See Figure 1: Typical Example of Thermal Strippers on page 7).

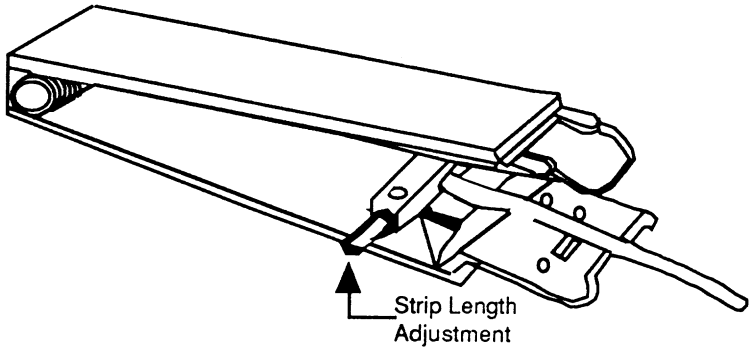


Figure 1: Typical Example of Thermal Strippers

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4.4. Holding Devices

Tools, fixtures, and material used to hold or restrain wires and components shall be of a design which will not damage or deform the wires, leads, wire insulation, or components. If toothed clips (alligator clips) are utilized, the jaw of the clips shall be covered to provide the required protection.

4.5. Bending Tools

Bending tools used for wire or lead bending may be automatic or hand implements and shall be of a material that will not cut, nick, or otherwise damage solid or stranded wires, leads, integral insulation, or any other insulation or insulators added prior to the bending operation. Bending tools shall be of a type that imparts no damage to the components bodies or seals.

4.6. Heat Source (i.e. Heat Guns)

4.6.1. Devices used for shrinking heat shrinkable tubing shall be temperature controlled to prevent damage to components or wiring.

4.6.2. Heat sources shall be maintained and kept clean in order to ensure uniform heat transfer and prevent contamination.

4.7. Soldering Equipment

4.7.1. Equipment shall be in accordance with DOD-STD-1686 for ESD protection.

4.7.2. Soldering irons and resistance heating elements shall heat the connection area rapidly and maintain proper soldering temperature at the connection throughout the soldering operation.

4.7.3. A soldering iron holder shall be of a type satisfactory for the soldering iron utilized. The holder shall leave the soldering iron element and tip unsupported without applied excess physical stress or heat sinking and shall protect personnel from burns.

4.7.4. Sponge pads for wipe cleaning the soldering iron tip shall be finely textured and sulfur free.

4.7.5. Solder pots shall be capable of maintaining the solder temperature at 260 ± 11 °C [500 ± 20 °F]. Solder pots shall be grounded.

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4.8. Crimping Tools

Crimping tools shall be controlled as follows to assure the reliability of crimped connections. Tensile tests are used to verify compatibility of the crimp device and matching tooling, and to establish initial calibration for each size and die combination used. Routine calibration gauging tests shall be used to determine that the crimping tool and die meets the estimated parameters.

a. Tensile Test - 2 to 3 sample specimens shall be fabricated for every different combination of crimp device, wire size and dies for each tool used. Wire retention capability of the crimped connection shall be tested using a tensile tester with an accuracy of 0.5% at a head travel of 1.27 mm [0.05 inch] per minute. The mechanical connection of the wire and crimp device shall not break or become distorted before the minimum tensile strength specified in Table 5: Acceptable Terminal Pull Test Value on page 18 (See Section 7.1. Method on page 18 of this specification) is reached.
b. Gauging Tests - each crimping tool shall be gauged prior to use, each day, using the tool manufacturers gauging procedures and parameters. Reset and recalibrate any tool which exhibits premature ratchet release before the dies mate (movement of dies is discernible) or where the dies are misaligned.
c. Operational Tests - operation of each crimping tool shall have a smooth operation with an additional force necessary to be exerted to the handles to tip the ratchet. Each tool which is found to have rough operation shall be serviced. Tools with faulty ratchet operation shall not be used. The crimped connection shall be easily removed from the tool upon completion of the crimping operation.
d. Capacity Study - for any new terminal introduction run capacity study for termination tooling and record CPK results.
e. Records - record results of gauging and tensile testing for each tool so that an individual tool history is maintained.

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4.9. Other Tools

a. Flush Cutting Pliers
b. Thermometer (Solder Pot)
c. Hand Vise
d. Labeled Flux Container
e. Solvent Container
f. Finger Cots and Protective Clothing
g. Stiff Bristle Brush (Anti-Static)
h. Metal Tweezers
i. Camel Hair Brush
j. Round Nose Pliers
k. Needle Nose Pliers
l. Dross Removal Tool (Stainless Steel Spoon)

5. Materials

If material is not specified on harness drawing, use material per following specifications.

5.1. Wire

5.1.1. All battery wire shall conform to SAE J1127.

5.1.2. All low tension Primary wire shall conform to SAE J1128.

5.1.3. All High temperature wires shall conform to MIL-W-16878, Wire and Cable Specification or ISO 6722-1, Road vehicles 60 V and 600 V single core cables, Part 1, Dimensions, test methods and requirements for copper conductor cables.

5.2. Covering

Convoluted conduit and braiding; when specified, materials shall meet the following performance criteria:

a. Maximum Operation Temperature: - 40 °C to + 125 °C continuous, + 150 °C intermittent.
b. Braiding material diameter not to exceed 0.71 ± 0.076 mm [0.028 ± 0.003 inches].

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5.3. Electrical Tape

Adhesive tape for bundling electrically conducting material shall be polyvinyl chloride (PVC) tape having a pressure sensitive adhesive coating on one side.

5.3.1. Tape shall meet the following properties and performance criteria:

a. Temperature Ranges - 40 °C + 125 °C
b. Total Thickness: 0.177 ± 0.025 mm [0.007 ± 0.001 inch]
c. Backing Thickness: 0.152 ± 0.025 mm [0.006 ± 0.001 inch]
d. Tensile Strength, Minimum: 3.5 KN/m width [240 lb., force/foot width]
e. Adhesion, Minimum: 219 N/m [15 lb. force/foot]
f. Dielectric Strength, Minimum: 10,000 VRMS at 60 Hz per SAE
g. Insulation Resistance, Minimum: 20,000 MΩ
h. Color: Black
i. Shall pass chemical spray test per CES 14232

5.3.2. All other tape used for covering break out and wire bundle shall meet environmental performance standards referenced in CES 14311.

5.4. Insulating Heat Shrinkable Sleeves

5.4.1. Sleeves shall be constructed of a heat shrinkable irradiated polyolefin with adhesive, to cover splice. Non-adhesive heat shrinkable black tubing is permitted for use only to ring and fork terminals and cover wire bundles. Preferred color for heat shrinkable tubing is black unless color-coding is needed for bundles or specific circuits and groups. Typical shrinkable ratio shall be 50% of a diameter when exposed to heat (149 °C).

5.5. Solder

Solder material shall comply with CES 10903.

5.6. Flux

Fluxes conforming to ROL0, ROL1, REL0 or REL1 of ANSI-J-STD-004 shall be used for making soldered connections.

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5.7. Solvents and Cleaners

5.7.1. The solvents or aqueous cleaners used for removal of grease, oil, dirt, flux, and other debris, shall be selected for their ability to remove both ionic and non-ionic contamination. The solvents or cleaners used shall not degrade the materials or parts being cleaned. A list of approved solvents and cleaners is provided in Table 1: Solvents on page 12 and Table 2: Cleaners on page 12. Mixtures of the approved solvents may be used.

Table 1: Solvents

Solvent	Specification
Ethyl alcohol	O-E-760, Types III, IV, V
Isopropyl Alcohol	TT-I-735

Table 2: Cleaners

Solvent	Specification
Water (Deionized)	1 MΩ-cm, minimum resistivity

5.7.2. Whenever deionized water is used, care shall be used to ensure that proper drying is accomplished immediately after its use.

5.7.3. Caution: Sonic or ultrasonic vibration shall not be used as a method for cleaning electrical or electronic parts and assemblies.

6. Wire Preparation

The first step in assembly is wire preparation. All wires shall be properly prepared prior to wire termination. Verification of proper wire preparation shall be by 100% inspection by operators and/or sample inspection by inspector in case of semi-automatic machines used for cutting and stripping of wires.

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6.1. Method

The method of preparation for general purpose wire is, wire identification, cutting, insulation removal (wire stripping), tinning (if required), and wire twisting (if required). The preparation requirements are as follows:

6.1.1. Wire Identification

Wire identification is not required. The manufacturer may use wire identification to support the wire harness assembly process if necessary.

6.1.1.1. Color Coding:

Refer to latest revision of EP00046857.

6.1.1.2. Wire Marking/Printing:

If wire marking identification is used, each wire in the harness should be permanently marked with its own (corresponding) circuit number. All marking media should contrast with wire insulation color (for legibility). The recommended placement of wire identification should be per Table 3: Wire Marking on page 13.

Table 3: Wire Marking

Wire Length	Identification Marking
up to 149 mm	Within 50 mm at each end.
150 mm to 599 mm	Within 75 mm at each end and every 150 mm. Tolerance ± 20 mm
600 mm to 2,499 mm	Within 75 mm at each end and every 300 mm. Tolerance ± 50 mm
2,500 mm and more	At 75 mm, 225 mm and 375 mm at both end. Tolerance ± 50 mm

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6.1.1.2. ....(Continued)

The marking shall have below properties:

a. Flame Retardant: Material that can withstand combustion and delay the passage of flame for a known time period.
b. Chemical Resistance: The printed information on the label shall remain legible for the life of the product.
c. High Temperature Resistance: The material shall withstand 125 °C continuous unless otherwise specified without degrading/melting/fading/peeling, etc.

6.1.2. Wire Cutting

The wire cutting operation requires the use of hand operated or automatic wire cutting machinery appropriate for the cutting of wires to required lengths as defined by Engineering Drawing.

6.1.3. Insulation Removal (Wire Stripping)

Wires shall be stripped of insulation per recommendations by terminal manufacturer. When a tolerance is not supplied by the terminal manufacturer, a tolerance of + 0.8 mm / - 0.5 mm [+ 0.032 inch / - 0.020 inch] shall be used. The general rule is that no stands are to be cut. However, a maximum of 6% stand cut or fewer is permitted (Example: one stand cut out of 19 stands, zero stands cut out of 7 stands). Manual stripping shall be completed by removing insulation with a smooth even motion in the direction of the wire lay.

Note: If wire lay is distributed after insulation removal, twist wire so as to restore original lay. (Bare fingers shall not be used to carry out this procedure/rework).

Note: In the event that thermal strippers are used, ensure that the thermal stripper contact fingers are clean. Application of too much squeezing pressure may damage underlying conductor surfaces (re-ringing). This is to be avoided.

6.1.4. Tinning (if required) is accomplished by the use of a calibrated solder pot. Remove the dross from the solder surface and dip the prepared stripped and fluxed wire end into the molten solder to the desired length to be tinned and allow the lead to remain in the solder for three to five seconds. Slowly rotate the lead as you remove it from the solder, completing the cycle within the five seconds.

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6.1.4. ....(Continued)

Tinned conductors shall meet the following requirements:

a. There shall be no bulging/swelling of insulation due to excessive wicking of solder.
b. Surface shall be free of all flux residues and contamination.
c. The tinned surface shall be continuous, smooth, without inclusion, and shall not show evidence of either non-wetting or dewetting.
d. Strands of stranded wire shall be visible under the solder coating and insulation shall not be damaged. (Wire strand contour shall be visible where the conductor enters the insulation).
e. Wire lay shall not be disturbed.

6.1.5. Twisted pair, triplet and quadruplet shall be consistent throughout the length of twisted wires. See Figure 2: Twisted Wire on page 15 for one twist definition. Twist wires per Table 4: Twisted Wire (Wire Diameters Per SAE J1128) on page 16.

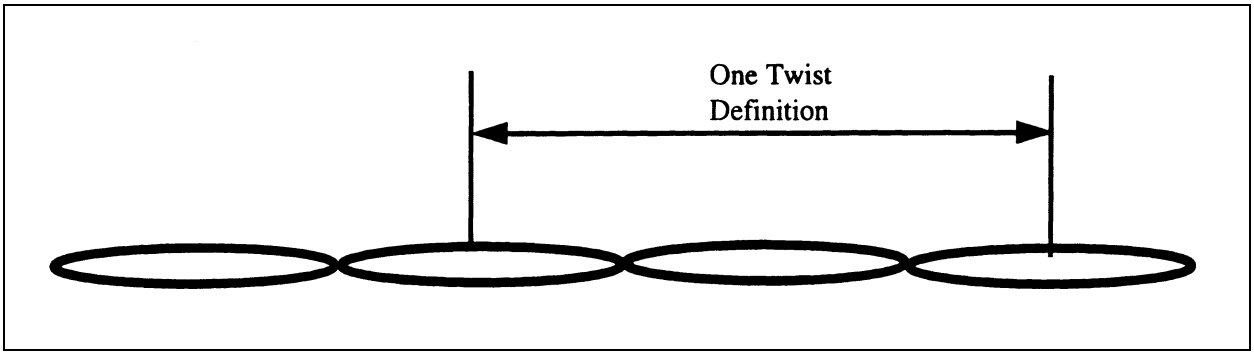


Figure 2: Twisted Wire

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Table 4: Twisted Wire (Wire Diameters Per SAE J1128)

Wire Size		O.D. of Cond. (In mm)			1 Twist Per Each mm Pairs or Triplets		
AWG	Metric	TXL	GXL	SXL	TXL	GXL	SXL
20	0.5	1.90	2.40	2.80	17	22	25
18	0.8	2.20	2.50	3.00	20	23	27
16	1.0	2.40	2.90	3.40	22	26	31
14	2.0	2.70	3.20	3.90	24	29	35
12	3.0	3.30	3.80	4.60	30	34	41

6.2. Special Cable

Special purpose cable preparation such as for SAE J1939 twisted pair cable and shielded cables require special preparation processes. When characteristic impedance or other circuit parameters are affected, such as high voltage circuits or coaxial line terminations, the insulation clearance requirements may be modified. All variations shall be documented in the assembly procedure. The following is recommended practices for special cable preparation per Figure 3: Shielded Cable Wire Preparation Process on page 17.

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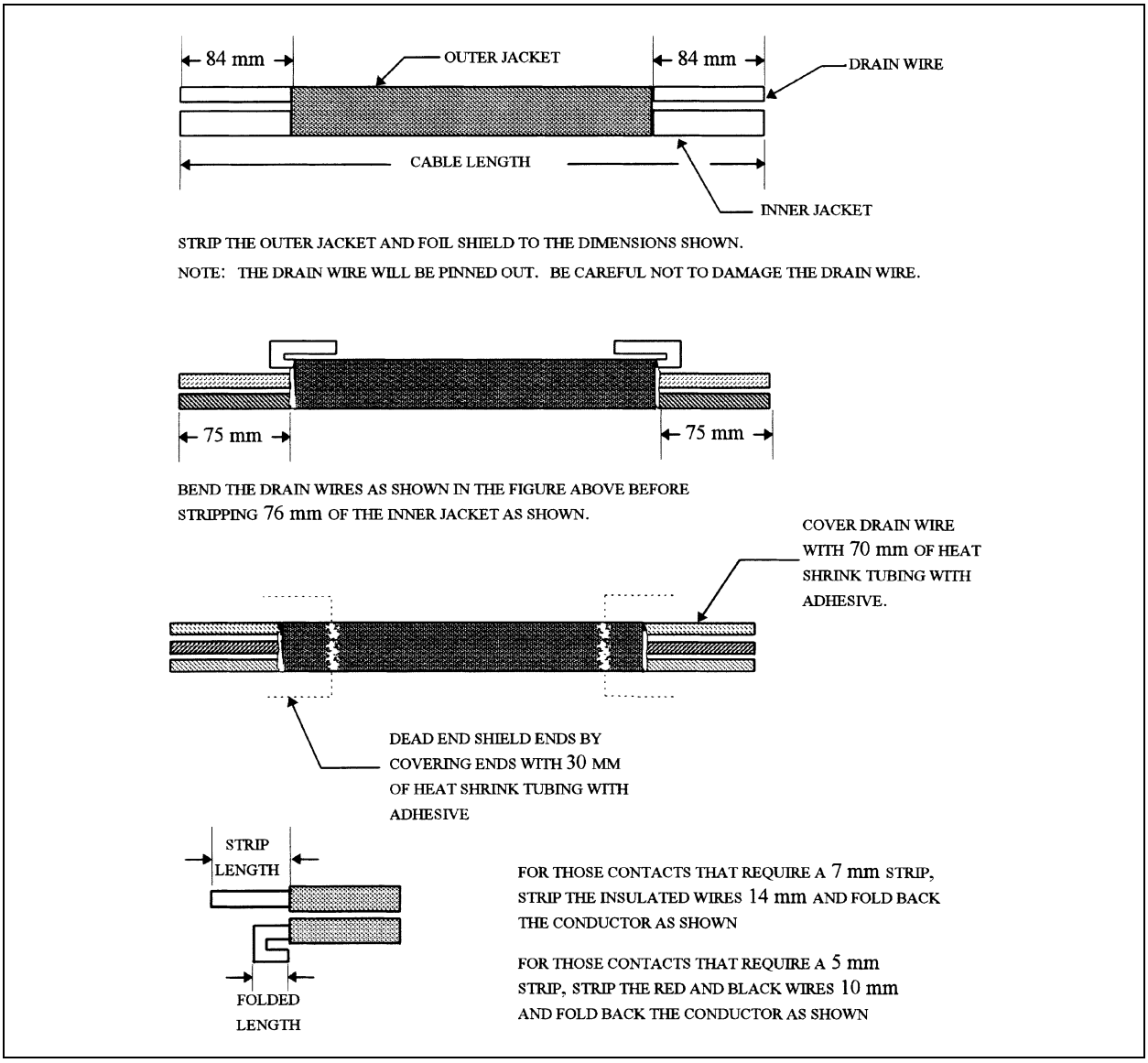


Figure 3: Shielded Cable Wire Preparation Process

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7. Crimping (Terminal to Wire)

7.1. Method

Crimping shall be done with the terminal manufacturer's recommended automatic or semi-automatic tooling. At no time shall a general purpose crimp tool be used in the manufacture of production harness. Each terminal pull value shall be checked after each setup and monitored during the production run. The test specimen terminal shall be crimped at Strands and also on the insulation. The insulation crimp shall be opened prior to conducting the pull test. If the terminal pull force is less than the value in Table 5: Acceptable Terminal Pull Test Value on page 18, but meets the manufacturer's recommended value, the tool may be used provided the terminal pull force value is within 15% of the table value. If the manufacturer's value is not within 15% of the value in the table, the tool may still be used after written approval from Cummins Engineering.

Table 5: Acceptable Terminal Pull Test Value

Size (mm^2)	Size (AWG)	Minimum Pull force (Pounds)
0.5	20	20
0.8	18	25
1.0	16	35
2.0	14	50
3.0	12	70
5.0	10	80
8.0	8	95

Note: No loose wire strands are permitted at the terminal; all wire strands shall be within the crimp.

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7.1.1. Mechanical Crimp

7.1.1.1. Prior to Crimping: All strands shall be enclosed within the crimp. No strands outside the crimp or broken-off before the crimp are allowed. Confirm the crimping operation does not affect the contact, locking, connector mating, or insertion functions of the terminal. Strands in stripped wire shall not be cut, missing, excessively nicked (significant strand reduction), or elongated. Insulation shall not be stuck or embedded in strands.

7.1.1.2. Cross-Section Analysis: Cross-sectional analysis is used as an aid in determining why a crimp passes or fails a portion of this test. Failure to pass an electrical test may be due to uneven strand dispersion, inadequate wing closure, voids, wings bottoming out, etc. Cross-section views of conductor crimps shall comply with Figure 4: Conductor Crimp Cross-Section Criteria on page 19.

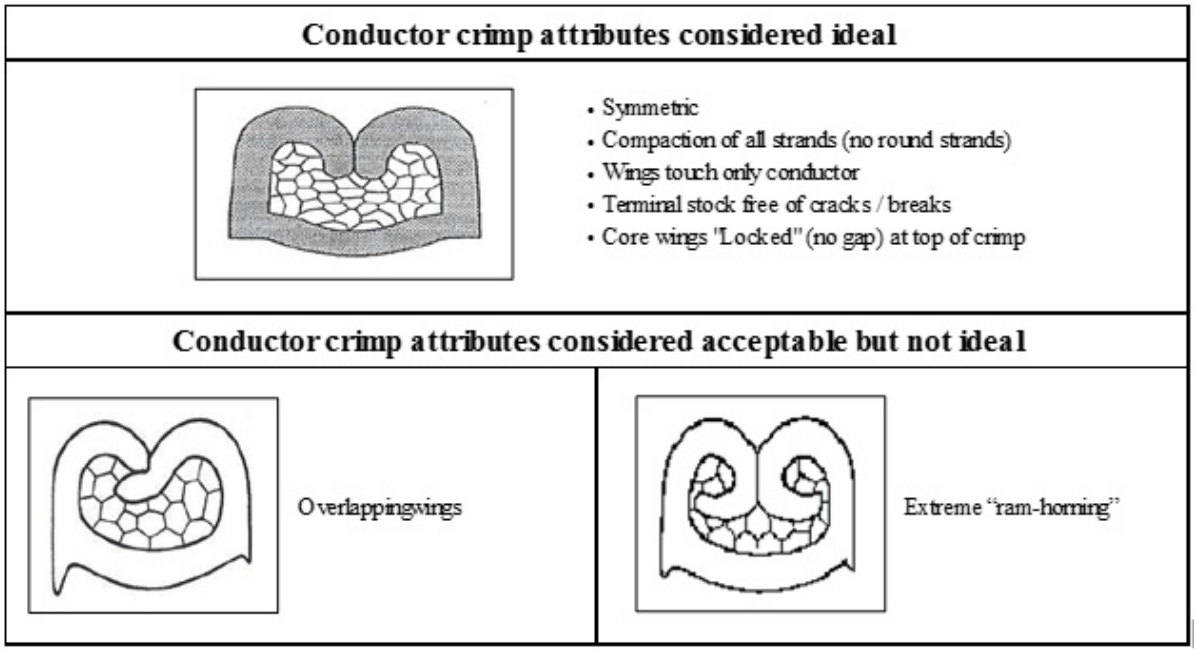


Figure 4: Conductor Crimp Cross-Section Criteria

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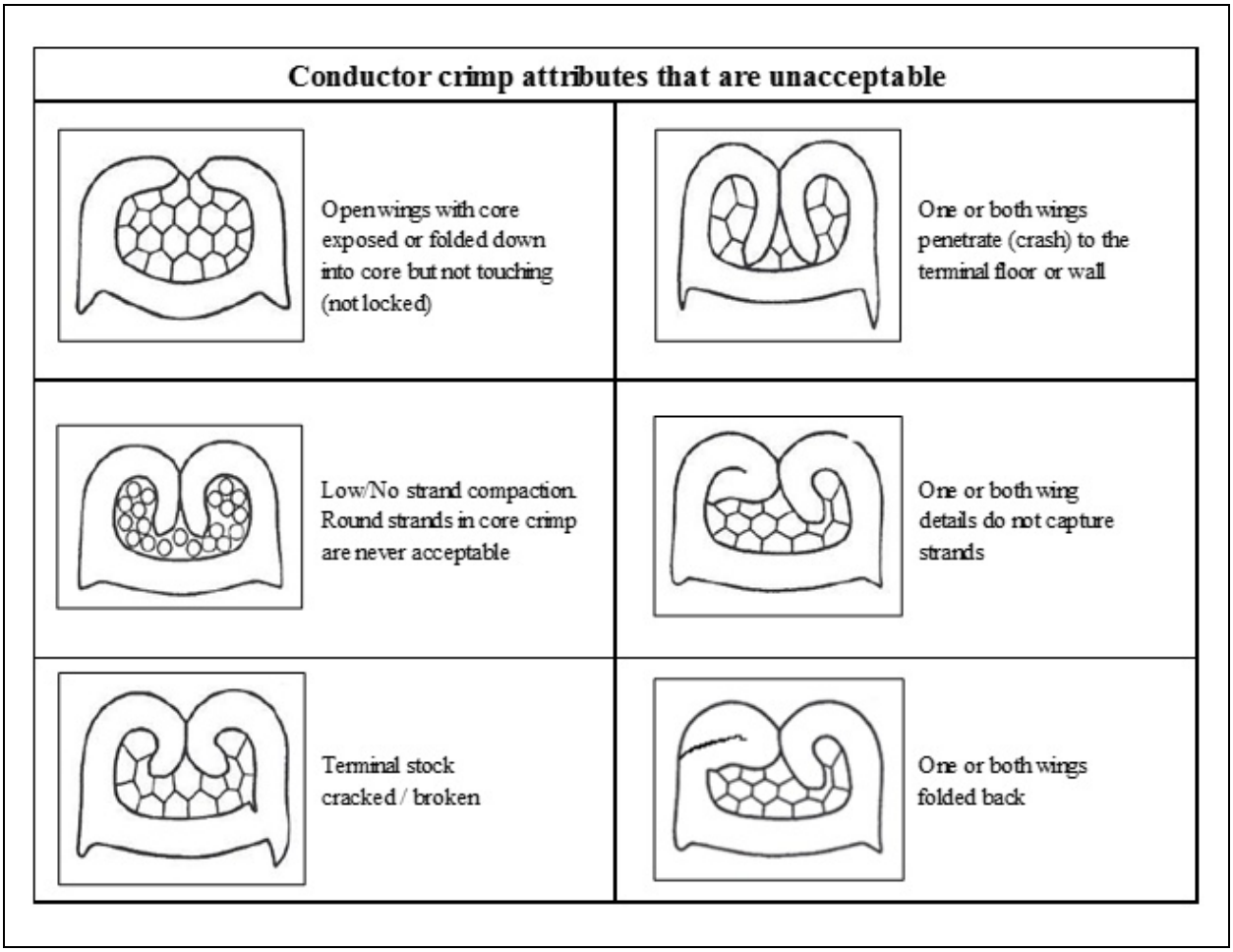


Figure 5: Conductor Crimp Cross-Section Criteria (Continued)

7.1.1.3. The width of the burr on the base of a crimp shall not exceed 0.1 mm for ≤ 0.8 mm thick terminal stock or 15% of stock thickness for > 0.8 mm stock. (For accurate crimp height measurement, it is recommended that the burr height not exceed the lowest position of the crimp.) The burr shall not cause damage to any subsequent operation. Special care shall be taken when using mat seals that can be cut by a burr.

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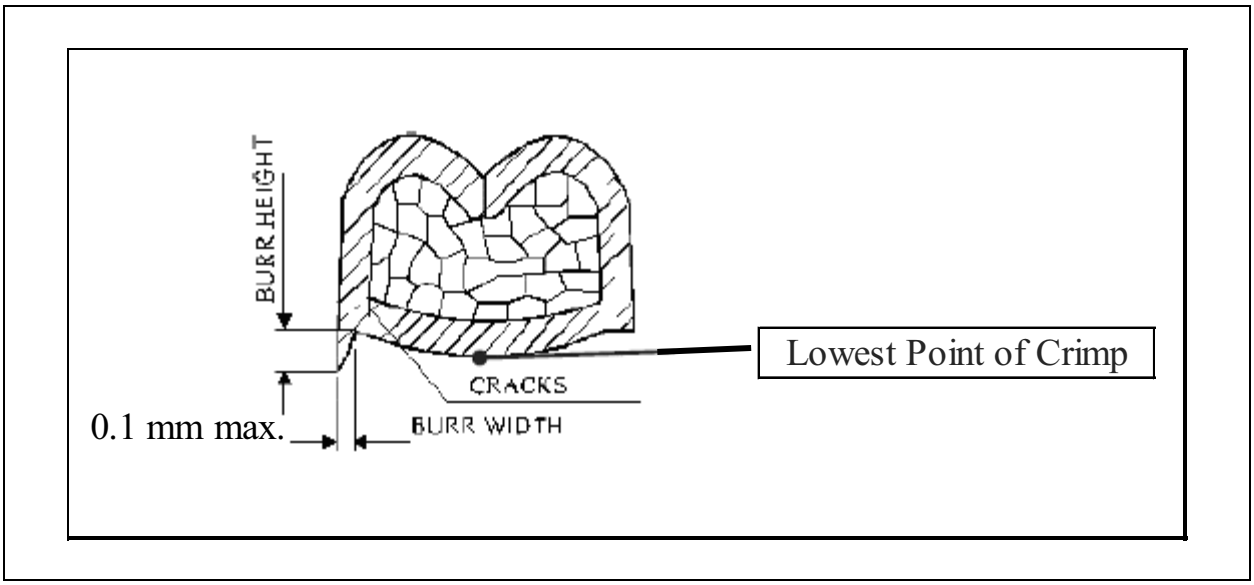


Figure 6: Conductor Crimp Burr Detail

7.2. Crimp Height

The core crimp height shall be per recommendations by the terminal manufacturer. When a tolerance is not supplied by the terminal manufacturer, the tolerance for 12 to 18 AWG wire shall be ± 0.10 mm [0.004 inch].

7.3. Crimping Tool Preventive Maintenance/Certification

The supplier shall maintain the crimping tools based on Terminal supplier's recommendation in terms of calibration/certification based on number of crimps or time interval, whichever comes first.

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7.4. Visual Aides

Use terminal manufacturer visual sample aid if supplied by manufacturer. The following inspection criteria shall be cause for rejection:

a. Burned, nicked, gouged or otherwise damage wire insulation.
b. Nicked or broken wire strands.
c. Scratched, gouged, nicked or otherwise damaged contact plating. The actual crimped area on a contact are not considered as contact damage.
d. Bent or broken contacts.
e. Bent contacts which have been straightened.
f. Bent insert tines or damaged contact retention clips.
g. Wire insulation tag ends.
h. Contacts which indicate double crimping.
i. Contacts which have been crimped by tools without a valid calibration decal.
j. Crimp indentation contacting either the inspection hole or the wire crimp barrel rim.

7.5. Rework-Wrong Cavity Terminal Insertion

a. Use terminal extraction tool recommended by connector manufacturer to remove terminal inserted in wrong cavity.
b. Be careful to ensure there is no damage to Seal and surrounding terminals are not bent or damaged during extraction of terminals.
c. It is also important to ensure no damage is done to the Terminal retention features in the cavity/on the terminal.

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8. Splice Construction (Wire to Wire)

Ultrasonic welding and Mechanical Crimp (round barrel type splices and brass splice clips) are used when two or more wires need to be joined. Ultrasonic Weld is the preferred method for splices. However some sensors require the ability to obtain an atmospheric pressure through wire to provide a reading relative to ambient (gauge pressure). To ensure proper sensor operation use mechanically crimped splices (if needed) for such circuits. Round barrel mechanical crimp splice can be used as an alternative method. The round barrel splice devices are proven as more reliable compared to the "C"- Clip type crimp splice devices, which are prone to failures for on engine application. All splices shall be insulated with sealing type double wall heat shrink tube. Heat shrink tube shall extend a minimum of 10 mm [0.4 inch] ± 2 mm [± 0.08 inch] beyond the splice joint over the cable insulation to effect an environmental seal. If more than one wire exits one end of a heat shrinkable sleeve, adhesive tubing shall be used to create an effective environmental seal. Splices shall be separated by a minimum of 18 mm [0.71 inch] from other splices, and located approximately as shown on the drawing. Splices shall not be used unless specifically indicated on the drawing.

8.1. Splicing Guidelines

8.1.1. Splices may be either mechanically crimped, ultrasonically welded, or resistance welded for plated wire. Experience has shown ultrasonic welds to be very consistent and of high quality, and are preferred to mechanical crimps and resistance welding for production assembly.

8.1.2. Splices should be staggered throughout the harness, so as not to create bulges, or irregular bundle diameters.

8.1.3. Splices shall not be located where significant harness flexing occurs, within 50 mm of a branch/breakout of the harness, within 100 mm of any connector or termination, and not in a drip loop.

8.1.4. Splices shall not be soldered as solder wicking through conductor can cause failures.

8.1.5. Splices should be sealed with an adhesive lined heat shrinkable material, specifically designed for harness splices, to form a gas tight seal. The heat shrinkable material shall be compatible with the fluids that may come in contact with the harness such as diesel fuel, oil, hydraulic fluids, coolant, and solvents. An overlap of 25 mm [0.98 inch] is recommended to provide adequate overlap of the heat shrink tubing over the wire insulation to ensure a gas tight seal. Processing time shall be sufficient to allow a good re-flow of the adhesive material around the splice. Sealed splices should be capable of withstanding a pressure differential of up to 5 psi without leakage.

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8.1.6. Semi-rigid heat shrink may be used to protect splices. Semi-rigid heat shrink protects splices by reducing bend moment stress and adds significant tensile strength to splice region.

8.1.7. For maintaining proper balance in splice maximum of 6 balanced wires in each splice shall be used. Allowable difference between numbers of wires in each side of a splice is 2 and maximum of 2 gage size difference between wires used in splice.

8.1.8. For cross section consistency check for ultrasonically welded splices with wires 3 gage sizes apart is shall. Please refer Unacceptable Weld bond cross section in Figure 9: Unacceptable Weld Bond Cross Sections on page 27. Cross section analysis of a good weld shall show that all strands are coherently welded.

8.2. Splicing Method

Crimping shall be done with splice manufacturer's recommended automatic or semi-automatic tooling. At no time shall a general purpose crimp tool be used in the manufacture of production harnesses. Each splice pull value shall be checked after each setup and monitored during the production run. If the splice pull force is less than the value in Table 5: Acceptable Terminal Pull Test Value on page 18, but meets the manufacturer's recommended value, the tool may be used provided the splice pull force value is within 15% of the table value. If the manufacturer's value is not within 15% of the value in the table, the tool may still be used after written approval from Cummins engineering. When multiple gauge wires spliced, use pull value for smaller gauge wire.

Splices should be configured so that wires are not doubled back within the harness and within the covered section of the harness.

8.2.1. Ultrasonic Welding

8.2.1.1. Acceptable weld bond appearance:

a. All wire strands welded.
b. All wire strands of the same length. Wire stands shall not be nicked or elongated.
c. Insulation shall not be stuck or imbedded in strands.
d. Strands shall be clean and free of lubricants and or discoloration.
e. Refer Figure 7: Acceptable Ultrasonic Weld Bond Appearance on page 25.

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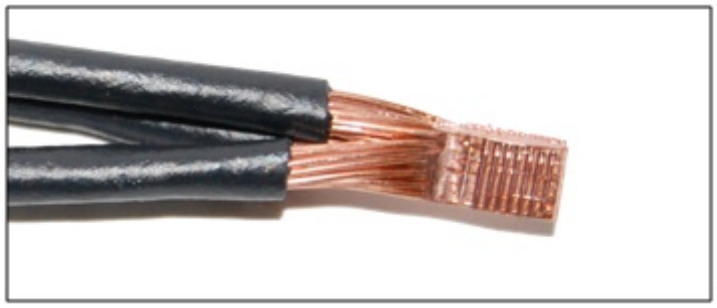


Figure 7: Acceptable Ultrasonic Weld Bond Appearance

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8.2.1.2. Non - Acceptable weld bond appearance:

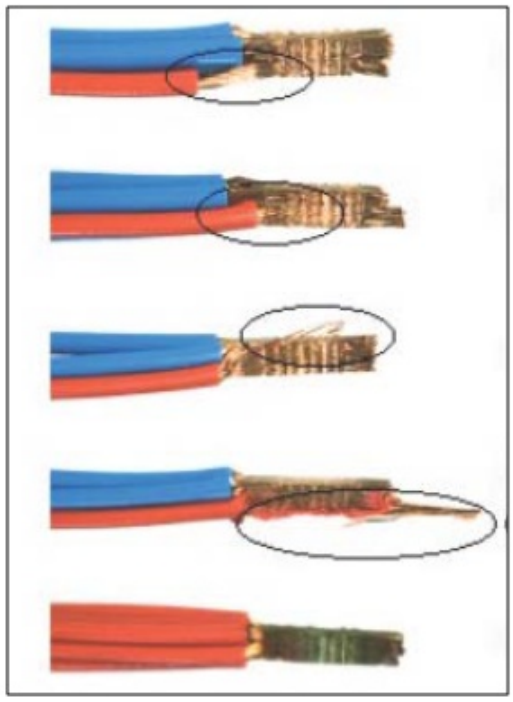


Figure 8: Non-Acceptable Ultrasonic Weld Bond Appearance

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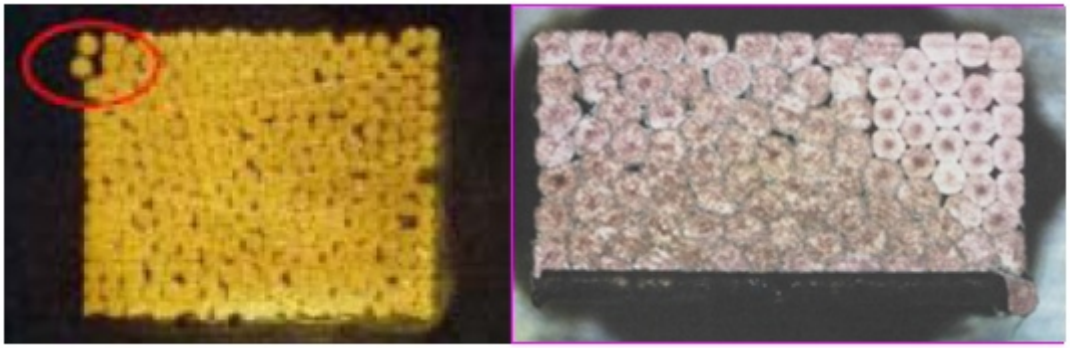


Figure 9: Unacceptable Weld Bond Cross Sections

8.2.2. Round barrel mechanical crimp splice:



Figure 10: Physical Appearance of Barrel Crimp

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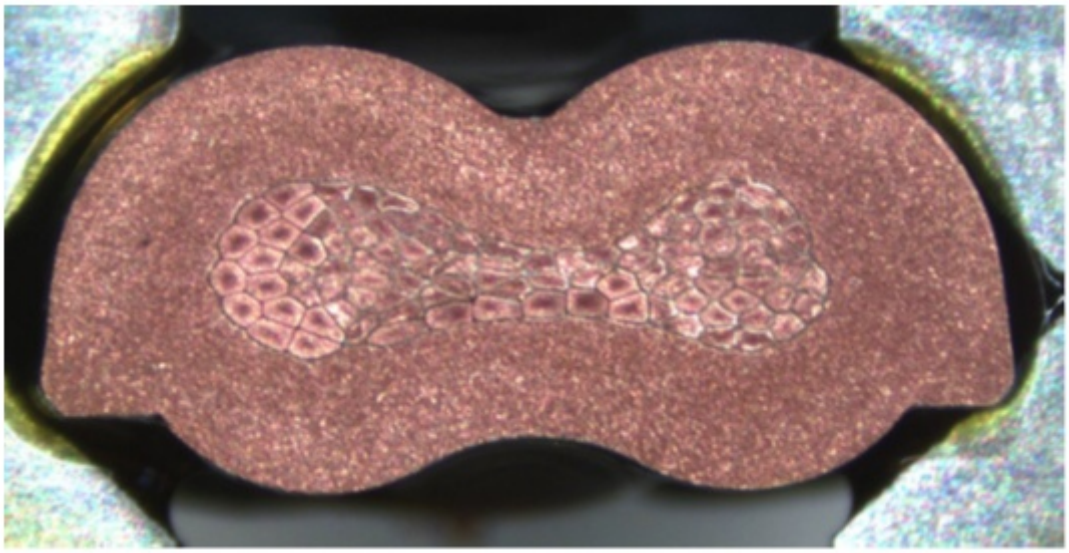


Figure 11: Acceptable Cross Section Micro Image of Round Barrel Crimp Splice

8.3. Adhesive Lined Heat Shrink Tubing Installation Method

a. Prior to applying heat shrink tube, splice shall be free from foreign material such as oil, tape, debris etc.
b. Heat the heat shrink tube until in contacts around the splice and wires.
c. Continue applying heat so that the adhesive flows and conforms to the areas between the wires. The entire outside area of heat shrink should be sealed. In addition the entire inside area of heat shrink should be sealed.
d. To verify the wire sealing, slice through the heat shrink, and wires, and look at wire cross-section (plus destructive test). No voids should be present in the adhesive as shown in Figure 12: Unacceptable Adhesive Tubing on page 29 and Figure 13: Acceptable Adhesive Tubing on page 29. There should be no gaps between the wires or heat shrinkable tubing.

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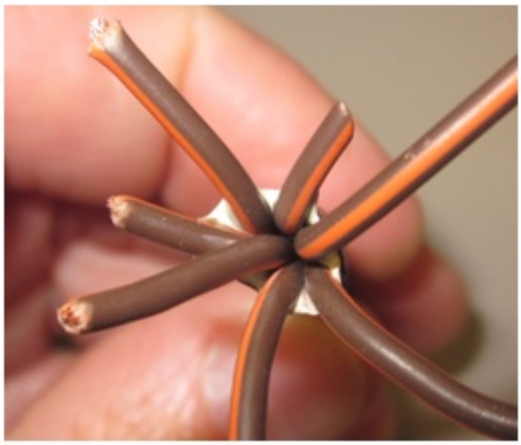


Figure 12: Unacceptable Adhesive Tubing

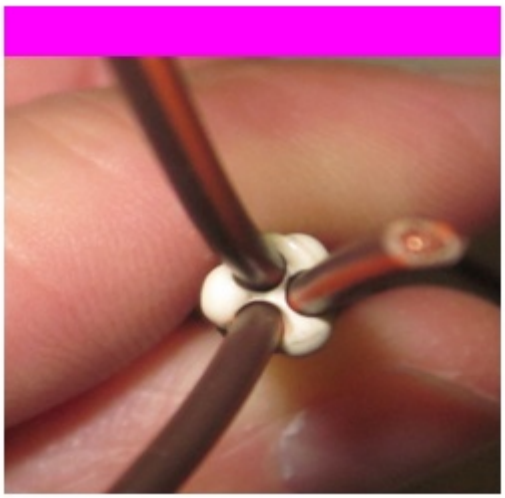




Figure 13: Acceptable Adhesive Tubing

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8.4. Alternative Terminal Splicing Method for Cables

This alternate method of cable splicing may be used if conventional splicing is physically restrictive. This splice should be used only when specified on the harness print. Also, it should only be used if the current is under one ampere. Figure 14: Construction Guide on page 30 should be used as a construction guide. Dimensions will vary given cable diameters, terminal type and connector construction.

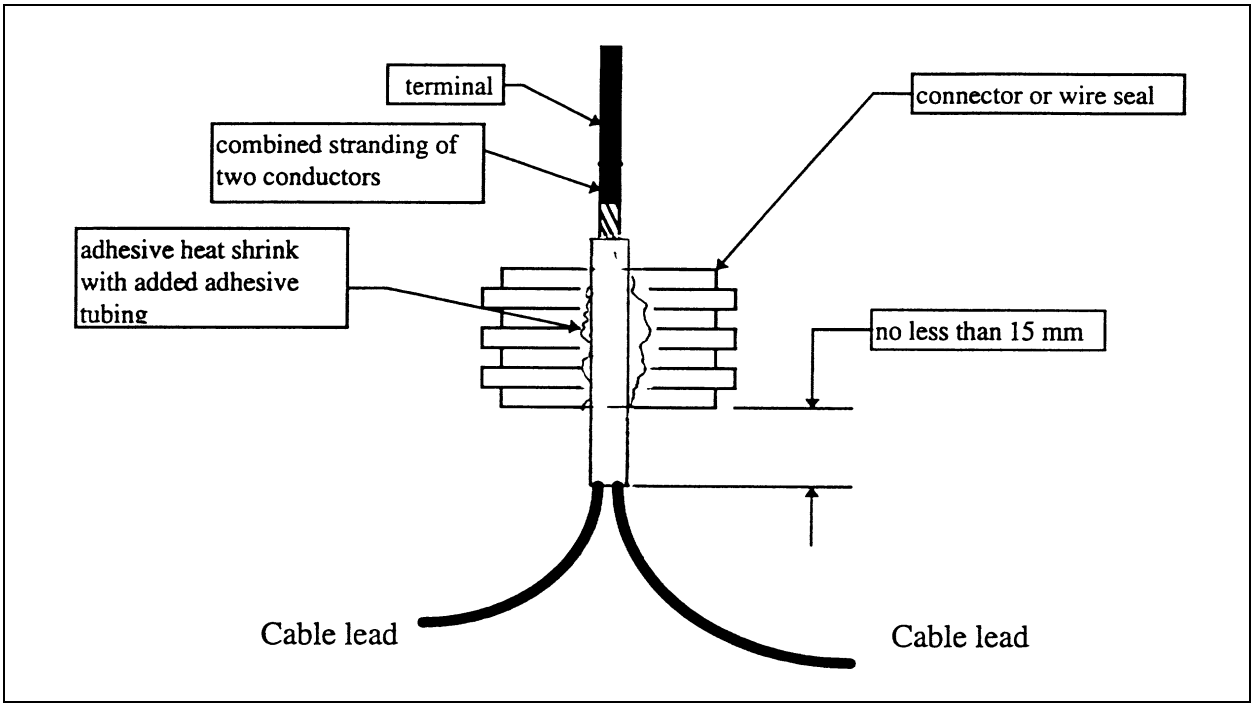


Figure 14: Construction Guide

9. Ring and Fork Terminals

All ring and fork terminals shall be sealed with adhesive heat shrink tubing. Apply 25 mm [0.98 inch] of general purpose heat shrinkable tubing at crimp area of the terminal. Tubing should not extend to mounting/interface area of terminals. Terminal pull test shall meet or exceed the minimum pull test as shown in Table 5: Acceptable Terminal Pull Test Value on page 18. Only copper terminals need to be tinned in a solder pot. Pre-tinned terminals do not require any soldering.

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10. Soldering

10.1. Function of Solder

The function of solder is to join two or more metals at temperature below their melting point. This joint provides metal and electrical continuity resulting in gas-tight, low-resistance connections. It also precludes corrosion at the junction. A solder joint also permits temperature-induced torsional stress without rupture of the joint.

10.2. Flux

The flux shall clean the metal but not attack, corrode or oxidize the metal. Acid type flux shall not be used.

10.3. Solder Joints

Acceptable solder joints shall have the following properties:

a. A smooth flow of solder to connecting metal.
b. No excess solder.
c. A bond between solder and metal.
d. No burned cable insulation.
e. No wicking or solder between the conductor and insulation crimps.
f. Solder shall remain in conductor crimp joint only.
g. No damage to spring members in the terminal.

10.4. Exclusions

Terminals shall not be soldered in the presence of nickel or gold plating in the area of the conductor crimp.

11. Connectors Preparation

11.1. Method

All wire lengths to a multi-cavity connector shall be cut and dressed to insure equal stress is on all wires after insertion into the connector. After insertion all terminals shall be checked for retention.

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11.2. Terminal Insertion

All terminal contacts shall be properly loaded into their respective connector shell cavities as specified by the drawing and per connector manufacturers specification to eliminate abnormal stress and deformation to terminals. All unused cavities shall be plugged with manufacturer cavity plugs.

For connectors without TPA (Terminal Position Assurance), use push-click-pull method to ensure terminal has seated properly.

11.3. Strain Relief

Allow uncovered wire at back of the wire seal, ring terminals and fork terminals for flexibility and strain relief. Use Table 6: Wire Flexibility Allowance on page 32 for allowable expose wire area:

Table 6: Wire Flexibility Allowance

Number of Cavities	Location of Allowable Exposed Wire	Allowable Exposed Wire
Connectors with less than 9 cavities	Dimension from back of connector to the beginning of the covering (i.e. convoluted tubing, braiding, tape, etc)	25 mm ± 5 mm
Connectors with 9 to 14 cavities	Dimension from back of connector to the beginning of the covering (i.e. convoluted tubing, braiding, tape, etc)	35 mm ± 5 mm
Connector with 14 to 24 cavities	Dimension from back of connector to the beginning of the covering (i.e. convoluted tubing, braiding, tape, etc)	50 mm ± 5 mm
Connector with 25 or more cavities	Dimension from back of connector to the beginning of the covering (i.e. convoluted tubing, braiding, tape, etc)	65 mm ± 10 mm

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11.4. Lubricant

11.4.1. As a standard Cummins practice, lubricant shall not be used on connector pins and sockets. A lubricant may be used only if the drawing explicitly calls for lubricant on specific connectors.

11.4.2. If lubricant is required, clean excess lubricant from connector housing. Note that the mating surface of connectors with female terminals shall be wiped dry after applying the lubricant, such that the lubricant is found only within the terminals themselves. Use lubricant per CES 20061.

12. Harness

12.1. Cable Ties

Plastic cable ties shall be used when specified on drawings.

12.2. Specification

The harness shall conform to the following drawings:

a. Circuit Data List
b. Assembly Drawing
c. Assembly Parts List
d. Drawing Notes

To ensure correct assembly and consistency between units, a 2 or 3 dimensional form board or mock-up build board simulating the interfacing modules can be used. Brackets to support the connectors and pins to guide the wires are recommended. All wires and connectors should be clearly identified to ensure correct termination and to facilitate continuity checking.

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12.3. Bundle Formation and Wire Grouping

As the wires are being formed and bound into a harness, the following requirements shall be taken into consideration:

a. The location of breakouts and splices; service loops and individual conductor length; and bundle shall result in the wires being uniformly dressed at the termination point (i.e. not taut wires at connectors).
b. Individual wires in a bundle shall lie parallel and not entwine other conductors except for “twisted wires” or branch leads crossing conductors.
c. Wires that are to break out together shall be grouped together.
d. Wire groups within a cable harness, approaching breakout points shall be positioned to minimize the amount of harness distortion.
e. There shall be no splices in the breakout areas.
f. Splices shall be located as near as practical to the source conductor termination whenever possible and buried within the bundle.
g. Splices on twisted wires shall be positioned for a maximum separation of 50 mm [1.9685 inches] between splices.
h. Splices contained in a cable harness shall be positioned in the harness to minimize the size of the bundle.

12.4. Taping

Spot taping shall be used to form and hold the wires in the harness; to secure convoluted tubing ends. Tape shall securely hold the bundle but not cut into the cable insulation.

12.5. Spiral

Tape spiral wrap entire cable using electrical tape except branches that are only a single twisted pair or a single twisted triple. See Figure 15: Spiral Wrap-Main Trunk and Branches on page 35.

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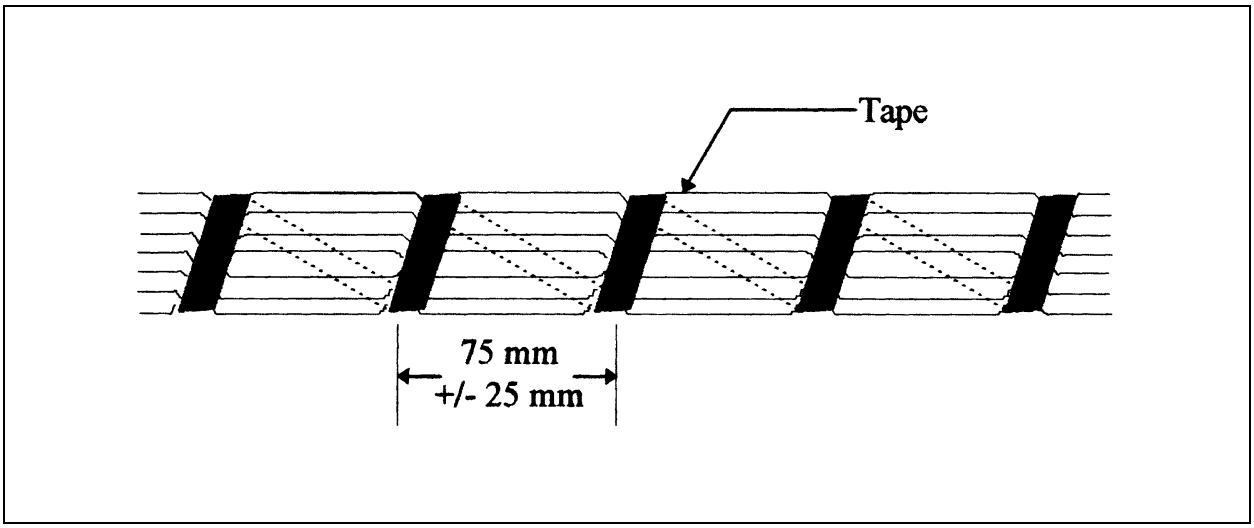


Figure 15: Spiral Wrap-Main Trunk and Branches

12.6. Manifolds (90° Elbow, Y-Joint, T-Joints) and Connector Backshell

12.6.1. "T" Joints shall be used only when specified on drawings. They shall be placed so that center leg or measurement reference point as specified by the designer, reflects the dimension referenced for the breakout.

12.6.2. Convoluted Conduit (Tubing)

Conduit may be either slit or non-slit. Non-slit is the design preference. Non-slit conduit helps prevent collapse of the conduit ends within the manifold or backshell when routed sharply, thus causing the conduit to pull apart. Also, non-slit conduit will not exhibit 'flaring' of the slit when the conduit is bent causing the inner bundle to be exposed or slip through the slit.

Conduit fill shall be as close to 75% as practicable. Cummins designers will select optimum sizing to prevent either difficulty in assembling non-slit conduit or prevent "rattling" of the bundle within an excessively large conduit.

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12.6.2. Convoluted Conduit (Tubing) (Continued)

Slit conduit shall be used in locations on the harness when a "J"-shaped downward loop is required by the engine harness routing design. Engine wash water or rainwater from the environment the engine will be operating within may not drain out fast enough around the connector backshell gaps and apertures. This is to prevent water intrusion issues through the wire-to-seal interfaces in connectors located lower than the uppermost end of the conduit "J"-shape.

Conduit sections when prepared to be assembled manifolds and backshell shall:

a. Have ends cut flush.
b. When slit conduit used, position the slit down or up depending on the harness bends, or as specified on the harness print. This prevents flare or unwanted opening of the slit which exposes the wire bundle or backside of the connector to debris, excessive liquids or potential physical damage.

12.6.3. Manifold and backshell assembly practices shall conform to the following:

a. Place minimum of three convolute ribs into the manifold or backshell or as specified by the manifold or backshell component manufacturer.
b. After manifolds and Connector backshell are closed, all latching tabs shall be completely engaged. Ensure all latches are undamaged after tab insertion as manifold and Connector backshell may open inadvertently.
c. When using PUR tubing, Nomex® wrap, or other similarly pliable harness covering, use miniature wire ties at the terminating end of such covering to anchor the covering underneath the respective manifold or backshell. Other methods such as single- or double-wall heat shrinkable tubing, or another method specified by the Cummins Inc. harness designer or recommended by the harness supplier may also be used as an anchor.

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13. Connector Potting

Connector potting is to provide strain relief at the junction of the connection pin and wire termination. This is required only when specified on the appropriate assembly drawing.

Sealed and cured connections shall conform to the following:

a. An electrical continuity test and solder joint inspection shall be performed prior to sealing.
b. The cured sealant shall be free of visible voids, bubbles, debris and shall not be tacky.
c. Sealant shall cover the tops of all solder cups.
d. Sealant applied over insulation sleeving shall coat any exposed areas of the solder cups including unused contacts and filled to the shrink tubing. Shrink tubing shall not be discolored, charred or split.

Where possible, all splices and terminations of cables should be included within the mold.

There shall be no potting material in any of the connector wires outside the boot and all floating connector contacts shall not be locked in place by the potting material.

14. Harness Identification

Finished harness assemblies shall be individually tagged to indicated Cummins Engine part number, manufacturer name or initials, date code (year/week; e.g. 92/27), and name of Country where harness is produced. This tag shall be an imprinted vinyl tape unless otherwise specified and shall be located as indicated on the drawing.

2345678	Harness Part Number (Cummins)
Rev A	Revision Level per Cummins print
Cummins Inc.	Cummins Name
Assembled in USA	Country of Assembly
9528	Date Code (1995, week 28)
ABC	Harness Manufacturer Name or Initials
	Harness Manufacturer ID code

Where applicable, for barcode specifications please refer CES 18287. Cummins Inc. harness designer shall specify requirement on harness drawing.

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15. Wire Harness Acceptance Criteria

15.1. Electrical Testing

All electrical testing shall be done by using spring loaded pogo pins. Preventative maintenance of test board shall be performed on regular schedule as per harness manufacturer's Process Control Plan.

15.2. Open Circuit, Short Circuit Test and Continuity

15.2.1. Harnesses with more than two circuits or with a multi-cavity connector shall be 100% electrically scanned for open circuits, short circuit and continuity by using a cable scanner. Continuity tester for point to point check is allowed for prototype and sample harnesses. Continuity tester is recommended to be a milliohmmeter with a 200-to-500 mΩ scale and measurement resolution of 0.1-to-0.5 mΩ, or equivalent as agreed upon by Cummins Inc. harness designer. All harnesses checked without using scanner shall be marked with yellow 12 mm [0.4724 inch] vinyl tape within 300 mm [11.811 inches] of main connector.

15.2.2. All repaired harnesses shall be scanned 100% after repair.

16. Foreign Material

All wiring harnesses shall be:

a. Free from dust, dirt, grease and other foreign debris.
b. Free from conducting particles, solder chips, conductor stands, metallic chips, etc., on surfaces, trapped in bundles and visible under insulation tubing.
c. Free of contaminants and solvent cleaning residues.

17. Packaging

All packaging shall meet the following criteria:

a. All packaging design and materials shall be agreed upon by the Cummins Inc. receiving plant or facility Packaging Designer, or equivalent staff. It shall comply with CES 19041.
b. All packaging shall conform to agreed international packaging design standards chosen by agreement between shipper and receiving plant or Cummins Inc. facility Packaging Designer or equivalent.

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17. Packaging (Continued)

c. When physically large harnesses are shipped, recommendation is to place each harness in its own individual bag per agreement by Cummins Inc. Plant Manufacturing Engineering/ Industrial Engineering staff. This prevents entanglement of branches and components and potential damage to harness or delays during short assembly station cycles. Ensure Shipping Caps are added to Connectors as per print to prevent damage to Terminals.
d. Large brackets and subassembly enclosures, when an integral part of the harness, shall be covered with padded wrap or other protective material per agreement with Cummins Inc. plant Packaging Designer or equivalent. This prevents physical damage to any part of the harness resulting from sharp edges found on brackets or enclosures.
e. All boxes should have plastic bag liner to protect harnesses from foreign material.
f. Pallets shipped by sea freight are recommended to be over wrapped with plastic to prevent excessive moisture condensation upon contents. Marine shipping practices should be understood.
g. Total weight of box or carton should not exceed 16 Kgm [35 pounds] or weight limit set by the Cummins Inc. plant or facility where harnesses are received.

18. Reference Readings

The following references were used in the preparation of this standard.

a. ANSI-J-Std-005, Requirements for Soldering Pastes
b. ANSI-J-Std-006, Requirements for Electronic Grade Solder Alloys and Fluxed and Non-Fluxed Solid Solders for Electronic Soldering Applications
c. CES 98187, Wiring Harness
d. CES 98408, Connector, Electrical
e. EP00043963, Wiring Harness Model Creation
f. EP00046857, Requirement for Wire Color in on Engine Harnesses
g. MIL-STD-1686, Electrostatic Discharge Control Program For Protection Of Electrical And Electronic Parts, Assemblies And Equipment (Excluding Electrically Initiated Explosive Devices)
h. SAE J473, Solders
i. USCAR21 Revision 3, Performance Specification for Cable-To-Terminal Electrical Crimps
j. USCAR38 Revision 0, Performance Specification for Ultrasonically Welded Wire/Cable Termination

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